CHAPTER EXERCISE: Answers for this chapter start on page 282.

A calculator should NOT be used on the following questions.

1

The ratio of a to b is 7:6, and the ratio of b to c is 8:5. If $a = 28$, what is the value of c ?

2

The ratio $2\frac{1}{4} : 1\frac{1}{2}$ can be written as $n : 2$. What is the value of n ?

3

The price of Product X is 25% greater than the price of Product Y. The price of Product Z is 25% less than the price of product Y. What is the ratio of the price of Product X to the price of Product Z?

A) 3:2
B) 4:3
C) 5:2
D) 5:3

A calculator is allowed on the following questions.

4

$$P = \frac{V^2}{R}$$

Electric power P is related to the voltage V and resistance R by the formula above. If the voltage were halved, how would the electric power be affected?

A) The electric power would be 4 times greater.
B) The electric power would be 2 times greater.
C) The electric power would be halved.
D) The electric power would be a quarter of what it was.

5

Julie has a square fence that encloses her garden. She decides to expand her garden by making each side of the fence 10 percent longer. After this expansion, the area of Julie's garden will have increased by what percent?

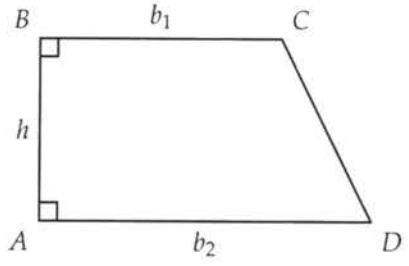
A) 20%
B) 21%
C) 22%
D) 25%

6

A right circular cone has a base radius of r and a height of h. If the radius is decreased by 20 percent and the height is increased by 10 percent, which of the following is the resulting percent change in the volume of the cone?

A) 10% decrease
B) 12% decrease
C) 18.4% decrease
D) 29.6% decrease

7



The area of the trapezoid above can be found using the formula $\frac{1}{2}(b_1 + b_2)h$. If lengths BC and AD are halved and the height is doubled, how would the area of the trapezoid change?

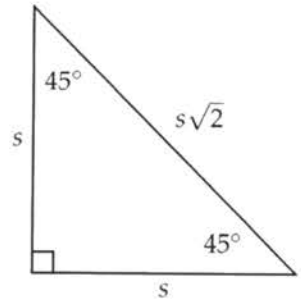
A) The area would be increased by 50 percent.
B) The area would stay the same.
C) The area would be decreased by 25 percent.
D) The area would be decreased by 50 percent.

8

Calvin has a sphere that is four times bigger than the one Kevin has in terms of volume. The radius of Calvin's sphere is how many times greater in length than the radius of Kevin's sphere (rounded to the nearest hundredth)?

A) 1.44
B) 1.59
C) 1.67
D) 2.00

9



In the triangle above, the lengths of the sides relate to one another as shown. If a new triangle is created by decreasing s such that the area of the new triangle is 64 percent of the original area, s must have been decreased by what percent?

A) 8%
B) 20%
C) 25%
D) 30%

Questions 10-11 refer to the following information.

$$L = 4\pi d^2 b$$

The total amount of energy emitted by a star each second is called its luminosity L, which is related to d, its distance (meters) away from Earth, and b, its brightness measured in watts per square meter, by the formula above.

10

If one star is three times as far away from Earth as another, and twice as bright, its luminosity is how many times greater than that of the other star?

A) 8
B) 9
C) 16
D) 18

11

Astronomers see two equally bright stars, Star A and Star B, in the night sky, but the luminosity of Star A is one-ninth the luminosity of Star B. The distance of Star A from Earth is what fraction of the distance of Star B from Earth?

A) $\frac{1}{27}$

B) $\frac{1}{9}$

C) $\frac{1}{3}$

D) $\frac{2}{3}$

▲

12

The student body at an after-school program consists only of 6th graders, 7th graders, and 8th graders. The ratio of 6th graders to 8th graders is 17 : 28. If a total of 110 students attend the program, n of whom are 7th graders, what is a possible value of n ?

13

A bookstore ordered an initial shipment of 10 paperback copies and 4 hardcover copies of a newly published book. The store must order a second shipment with the same ratio of paperback and hardcover copies as the initial shipment. If the store orders 50 hardcover copies of the book for the second shipment, how many paperback copies should the store order?

14

If the ratio of $y : 2.4$ is equivalent to $2.7 : 3.6$, what is the value of y ?

A) $\frac{3}{2}$

B) $\frac{4}{3}$

C) $\frac{7}{3}$

D) $\frac{9}{5}$

15

Box A weighs 42 pounds and Box B weighs 30 pounds. The ratio of the weights of Box A to Box B is equal to the ratio of the weights of Box C to Box D. If Box C and Box D weigh a total of 180 pounds, what is the weight of Box C, in pounds?

A) 50

B) 75

C) 105

D) 130

Chapter 5: Ratio & Proportion

CHAPTER EXERCISE:

1. [15] Since $a = 28$, the value of b is $\frac{6}{7} \times 28 = 24$. The value of c is then $\frac{5}{8} \times 24 = 15$.

2. [3] Remember that ratios are essentially just fractions. So the given ratio is equivalent to

$$2\frac{1}{4} \div 1\frac{1}{2} = \frac{9}{4} \div \frac{3}{2} = \frac{9}{4} \times \frac{2}{3} = \frac{3}{2} = 3:2$$

Therefore, $n = 3$.

3. [D] Let y be the price of Product Y. Then the price of Product X is $1.25y$ and the price of Product Z is $0.75y$. Simplifying the ratio of their prices, we get

$$\frac{1.25y}{0.75y} = \frac{1.25}{0.75} = \frac{5}{4} \div \frac{3}{4} = \frac{5}{4} \times \frac{4}{3} = \frac{5}{3} = 5:3$$

4. [D]

$$P_{old} = \frac{V^2}{R}$$

$$P_{new} = \frac{(0.5V)^2}{R} = \frac{0.25V^2}{R} = 0.25P_{old}$$

The electric power drops to a fourth of what it was.

5. [B] The area of a square is $A = s^2$, where s is the length of each side.

$$A_{new} = (1.10s)^2 = (1.10)^2 s^2 = 1.21s^2 = 1.21A_{old}$$

The new area is 21% greater.

6. [D]

$$V_{old} = \frac{1}{3}\pi r^2 h$$

$$V_{new} = \frac{1}{3}\pi(0.80r)^2(1.10h) = (0.80)^2(1.10)\left(\frac{1}{3}\pi r^2 h\right) = 0.704V_{old}$$

The volume of the cone decreases by $1 - 0.704 = 0.296 = 29.6\%$.

7. [B]

$$A_{old} = \frac{1}{2}(b_1 + b_2)h$$

$$A_{new} = \frac{1}{2}\left(\frac{1}{2}b_1 + \frac{1}{2}b_2\right)(2h) = \left(\frac{1}{2}\right)(2)\left[\frac{1}{2}(b_1 + b_2)h\right] = \frac{1}{2}(b_1 + b_2)h = A_{old}$$

Notice how $\frac{1}{2}$ was factored out from b_1 and b_2. The area stays the same.

8. [B] Let r be the radius of Kevin's sphere, and let x be the factor the radius of Calvin's sphere is greater by.

$$V_{Kevin} = \frac{4}{3}\pi r^3$$

$$V_{Calvin} = \frac{4}{3}\pi(xr)^3 = x^3\left(\frac{4}{3}\pi r^3\right) = x^3 V_{Kevin}$$

$$x^3 = 4$$

$$x = \sqrt[3]{4} \approx 1.59$$

9. $\boxed{B}$ The area of the original triangle is $\frac{1}{2}(s)(s) = \frac{1}{2}s^2$

$$A_{new} = \frac{1}{2}(xs)^2 = x^2\left(\frac{1}{2}s^2\right) = x^2 A_{old}$$

$$x^2 = 0.64$$

$$x = .80$$

s must have been decreased by $1 - .80 = 0.20 = 20\%$.

10. $\boxed{D}$

$$\begin{aligned} L_{other\ star} &= 4\pi d^2 b \\ L_{star} &= 4\pi(3d)^2(2b) \\ &= (3)^2(2)(4\pi d^2 b) \\ &= 18L_{other\ star} \end{aligned}$$

11. $\boxed{C}$ Let x be the fraction that Star A's distance is of Star B's.

$$L_{StarA} = \frac{1}{9}L_{StarB}$$

$$4\pi(xd)^2 b = \frac{1}{9}(4\pi d^2 b)$$

$$x^2(4\pi d^2 b) = \frac{1}{9}(4\pi d^2 b)$$

$$x^2 = \frac{1}{9}$$

$$x = \frac{1}{3}$$

12. $\boxed{20 \text{ or } 65}$ The given ratio means that there are 17 sixth graders and 28 eighth graders for every batch of $17 + 28 = 45$ sixth and eighth graders. This means that the total number of sixth and eighth graders must be a multiple of 45. In the case that this total is 45, the remaining $110 - 45 = 65$ students must be seventh graders. In the case that this total is $45 \times 2 = 90$, the remaining $110 - 90 = 20$ students must be seventh graders. Therefore, the possible values of n are 20 and 65. Notice that we don't have to consider multiples of 45 higher than 90 since those multiples would exceed the total number of students.

13. $\boxed{125}$ $\frac{10 \text{ paperback}}{4 \text{ hardcover}} \times 50 \text{ hardcover} = 125$ paperback copies

14. $\boxed{D}$

$$\frac{y}{2.4} = \frac{2.7}{3.6}$$

$$y = \frac{2.7}{3.6}(2.4) = 1.8 = \frac{9}{5}$$

15. $\boxed{C}$ The ratio of the weight of Box A to the weight of Box B reduces to 7 : 5. Since the weights of Boxes C and D follow the same ratio, Box C must weigh $\frac{7}{7+5} = \frac{7}{12}$ of the total weight. Therefore, Box C weighs $\frac{7}{12} \times 180 = 105$ pounds.